



Experiment 8

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Academic Year: 2025-2026

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Experiment 8: File Handling and Exception Handling

1. Add few names, one name in each row, in "name.txt file".
 - a. Count no of names
 - b. Count all names starting with vowel
 - c. Find longest name

```
In [ ]: # Open and read file
with open("emp.txt", "r") as file:
    names = file.readlines()

# Remove newline characters
names = [name.strip() for name in names]

# a) Count number of names
print("Total names:", len(names))

# b) Count names starting with vowel
vowels = ('A', 'E', 'I', 'O', 'U')
vowel_count = 0

for name in names:
    if name.upper().startswith(vowels):
        vowel_count += 1

print("Names starting with vowel:", vowel_count)

# c) Find Longest name
longest_name = names[0]

for name in names:
    if len(name) > len(longest_name):
        longest_name = name

print("Longest name:", longest_name)
```

Total names: 7

Names starting with vowel: 5

Longest name: Tshita

Longest name: Ishita

2. Store integers in a file.
 - a. Find the max number
 - b. Find average of all numbers
 - c. Count number of numbers greater than 100

```
In [ ]: with open("integer.txt", "r") as file:
        nums = [int(line.strip()) for line in file]

        # a) Maximum number
        max_num = nums[0]
        for num in nums:
            if num > max_num:
                max_num = num

        print("Maximum number:", max_num)

        # b) Average
        total = 0
        for num in nums:
            total += num

        average = total / len(nums)
        print("Average:", average)

        # c) Count numbers greater than 100
        count = 0
        for num in nums:
            if num > 100:
                count += 1

        print("Numbers greater than 100:", count)
```

Maximum number: 340

Average: 134.16666666666666

Numbers greater than 100: 3

3. Assume a file city.txt with details of 5 cities in given format (cityname population(in lakhs) area(in sq KM)):

Example:

Dehradun 5.78 308.20

Delhi 16.48 1484

Delhi 190 1484

.....

Open file city.txt and read to:

- Display details of all cities
- Display city names with population more than 10Lakhs
- Display sum of areas of all cities

```
In [ ]: with open("city.txt", "r") as file:
        cities = file.readlines()

        total_area = 0

        print("City Details:")
        for line in cities:
            city, population, area = line.split()
            population = float(population)
            area = float(area)

            # a) Display all city details
            print(city, population, area)

            # b) Population more than 10 Lakhs
            if population > 10:
                print("City with population > 10 lakhs:", city)

            # c) Sum of area
            total_area += area

        print("Total Area of all cities:", total_area)
```

City Details:

Dehradun 5.78 308.2

Delhi 190.0 1484.0

City with population > 10 lakhs: Delhi

Mumbai 204.0 603.4

City with population > 10 lakhs: Mumbai

Chandigarh 11.0 114.0

City with population > 10 lakhs: Chandigarh

Shimla 3.5 35.0

Total Area of all cities: 2544.6

4. Input two values from user where the first line contains N, the number of test cases. The next N lines contain the space

separated values of a and b. Perform integer division and print a/b. Handle exception in case of ZeroDivisionError or ValueError.

Sample input

1 0

2 \$

3 1

Sample Output:

Error Code: integer division or modulo by zero

Error Code: invalid literal for int() with base 10: '\$' 3

```
In [ ]: n = int(input("Enter number of test cases: "))

for i in range(n):
    try:
        a, b = input("Enter a and b: ").split()
        result = int(a) // int(b)
        print(result)

    except ZeroDivisionError as e:
        print("Error Code:", e)

    except ValueError as e:
        print("Error Code:", e)
```

Error Code: division by zero

Error Code: invalid literal for int() with base 10: '\$'

3

5. Create multiple suitable exceptions for a file handling program.

```
In [ ]: try:
        filename = input("Enter file name: ")

        with open(filename, "r") as file:
            data = file.read()
            print("File Content:\n", data)

    except FileNotFoundError:
        print("Error: File not found.")
```

```
except PermissionError:
    print("Error: Permission denied.")

except ValueError:
    print("Error: Invalid data format.")

except Exception as e:
    print("Unexpected Error:", e)

finally:
    print("Program Finished.")
```

Error: File not found.
Program Finished.